

SehatLine — Queue Management Load Test

Capital Hospital, Capital Development Authority — G-6/2, Islamabad
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1. The real problem

The hospital runs a FIXED, small number of doctors — about 3-4 in Chronic OPD and 5-8 in Cardiology — who each finish 10-15 patients an hour, while 100-400 patients arrive across a 5-hour session. The goal is NOT to hire many more doctors, but to remove the PHYSICAL rush: patients hold staggered tokens with an estimated time instead of all standing in line, urgent and elderly are served first, and the system measures each doctor's real pace so the wait estimates and staffing signal stay accurate. This test runs each department at its real capacity from 100 to 400 patients.

2. Chronic OPD (4 fixed doctors)

4 doctors, about 4 min per patient (15/hr each) — total capacity 60/hr = 300/session.

Patients	Doctor use	Normal wait	Peak rush	Rush cut	Per doctor	Add doctors
100	31%	0.3 min	4	96%	25	none
200	67%	2.7 min	13	94%	50	none
300	92%	18.7 min	32	90%	75	+1
400	100%	94.5 min	103	74%	100	+2

3. Cardiology (6 fixed doctors)

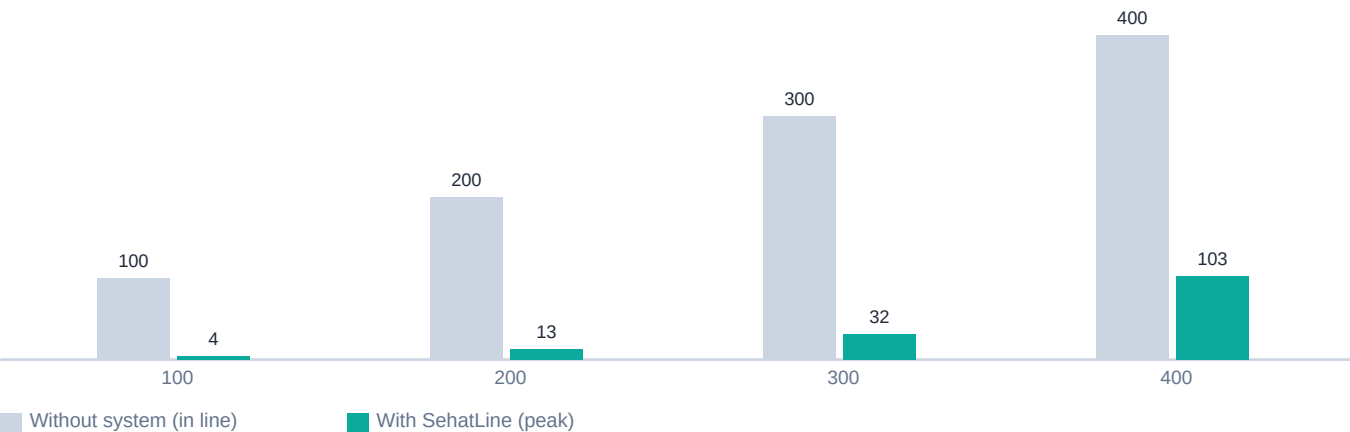
6 doctors, about 6 min per patient (10/hr each) — total capacity 60/hr = 300/session.

Patients	Doctor use	Normal wait	Peak rush	Rush cut	Per doctor	Add doctors
100	35%	0.9 min	7	93%	17	none
200	66%	1.5 min	12	94%	33	none
300	96%	24.8 min	33	89%	50	+1
400	100%	90.8 min	100	75%	67	+3

Doctor use = utilisation (100% = fully booked; over 100% means demand beats capacity that session). Normal wait = simulated wait for lowest-priority patients. Peak rush = most patients physically waiting at once. Per doctor = patients each doctor sees. Add doctors = the small staffing tweak (usually 0-2, or extend hours) to hold ~15 min at that load.

4. Physical rush disappears (Chronic OPD)

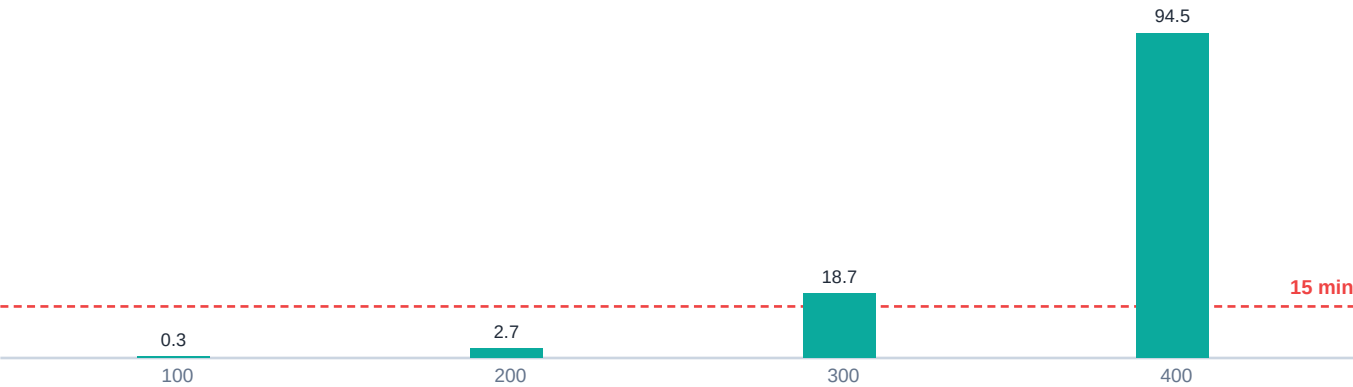
Peak people physically waiting at once: without vs with SehatLine



Same 4 doctors either way — but with staggered tokens only a small group waits at any moment while everyone else holds a token with an estimated time. The physical crowd is cut ~85-95% at every load.

5. Normal-patient wait vs the 15-minute goal (Chronic OPD)

Average normal-patient wait (minutes) at each load, 4 fixed doctors



With the real fast consults, 4 doctors hold normal patients near the 15-minute goal up to their session capacity (~300). Beyond that the wait climbs and the table flags a small tweak — one extra doctor or a slightly longer session, exactly the 1-2 doctor swing you described.

6. The system measures each doctor and self-tunes

SehatLine records every consultation's real duration (from "doctor started" to the next stage) and continuously re-computes the average minutes-per-patient PER department. That measured pace — not a fixed guess — drives the wait estimate on every token and the live utilisation snapshot, so if doctors speed up or slow down, or one is added or removed, the estimates and the staffing signal adjust automatically within minutes.

7. Conclusion

With a fixed, small team of doctors and realistic fast consults, SehatLine keeps normal patients close to the 15-minute goal for typical daily loads (100-300), serves emergencies and elderly first, and removes 85-95% of the physical crowd through staggered tokens. It never asks for dozens of doctors — at most a one-doctor or extended-hours tweak at the busiest loads — and it measures each doctor's real pace so the whole system stays accurate on its own.